

PETERSBURG, AK, USA

WMO: 703860

Lat: 56.806N

Lon: 132.937W

Elev: 33

StdP: 100.93

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25329

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-10.9	-17.3	0.8	-10.2	-14.1	1.1	-8.6	7.8	4.7	7.0	4.4	1.3	250	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	21.1	14.6	18.9	13.9	17.5	13.1	15.3	18.9	14.6	17.9	13.8	16.8	2.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	9.8	16.2	12.8	9.3	15.4	12.4	9.0	15.0	42.5	19.1	40.6	17.7	38.6	16.7	18.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.6	5.4	4.6	-15.7	24.7	2.2	2.2	-17.3	26.3	-18.5	27.6	-19.8	28.9	-21.3	30.5
			-15.8	16.6	2.2	0.9	-17.4	17.3	-18.7	17.8	-20.0	18.2	-21.6	18.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-0.2	0.4	1.5	4.9	8.8	11.8	13.4	13.1	10.3	6.4	2.2	0.1
	DBStd	5.97	4.97	3.92	3.42	2.23	2.69	2.27	1.80	1.80	1.89	2.51	3.79	4.51
	HDD10.0	1741	317	269	264	154	56	5	1	1	18	116	233	307
	HDD18.3	4471	575	503	523	404	296	196	152	162	241	372	483	566
	CDD10.0	312	0	0	0	1	19	60	107	97	27	2	0	0
	CDD18.3	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH23.3	8	0	0	0	0	0	5	2	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	1.6	1.5	1.5	1.8	1.9	1.9	1.7	1.4	1.3	1.4	1.6	1.6	1.6
Precipitation	PrecAvg	2561	229	185	182	135	117	99	124	161	317	345	294	256
	PrecMax	3256	472	373	378	246	332	277	224	404	506	561	556	563
	PrecMin	1428	24	56	22	10	0	16	41	50	172	33	71	42
	PrecStd	410	104	84	85	62	76	56	52	85	104	116	121	138
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	10.2	11.2	15.9	20.6	23.7	22.6	22.3	17.6	14.2	12.4	11.1
		MCWB	5.7	5.5	6.8	8.6	12.6	14.6	15.6	15.0	12.6	10.3	7.2	5.5
	2%	DB	8.0	8.0	8.2	12.4	17.6	20.1	20.9	20.0	16.0	12.5	9.2	8.7
		MCWB	5.5	4.8	5.2	7.2	11.4	13.6	15.1	14.6	12.0	10.0	7.2	6.2
	5%	DB	7.2	7.2	7.2	10.9	15.9	17.7	18.6	18.0	14.1	11.2	7.8	7.4
		MCWB	5.0	4.5	4.6	6.5	10.6	12.8	14.4	13.8	11.2	9.2	6.0	5.5
	10%	DB	5.9	6.0	6.1	8.9	13.7	16.2	17.3	17.0	13.0	9.9	7.1	6.0
		MCWB	4.1	3.8	3.6	5.7	9.5	12.0	13.5	13.5	11.0	8.3	5.2	4.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	6.6	7.7	9.4	13.4	15.4	16.4	16.4	14.5	11.9	8.6	7.9
		MCDB	8.4	8.5	10.1	13.6	18.9	21.4	21.4	20.7	16.9	13.2	10.2	9.0
	2%	WB	6.0	5.3	5.8	7.9	12.0	14.2	15.4	15.2	13.0	10.4	7.6	6.7
		MCDB	7.8	7.6	8.0	11.1	16.4	18.8	19.1	18.3	14.9	11.9	8.9	8.1
	5%	WB	5.0	4.6	4.9	7.0	10.9	13.3	14.6	14.6	12.2	9.3	6.4	5.3
		MCDB	7.1	6.9	6.9	9.4	14.4	17.2	17.8	17.5	13.4	10.6	7.9	7.4
	10%	WB	4.1	3.8	4.0	6.2	10.2	12.4	13.9	13.8	11.6	8.3	5.3	4.3
		MCDB	5.9	5.8	5.8	8.5	13.2	15.6	16.7	16.4	12.8	9.3	7.0	6.2
Mean Daily Temperature Range	5% DB	MDBR	4.2	5.3	5.9	7.1	7.3	6.4	5.9	6.0	5.6	4.8	4.3	4.2
		MCDBR	5.5	6.3	7.2	10.3	11.2	10.3	9.9	9.4	7.2	6.5	5.4	5.5
		MCWBR	3.6	3.7	4.7	6.0	6.0	5.5	5.4	5.3	4.6	4.3	3.5	3.7
	5% WB	MCDBR	5.2	5.7	6.5	8.3	9.0	8.9	7.9	7.9	5.7	5.5	4.9	5.4
		MCWBR	3.6	3.7	4.5	5.4	5.2	5.0	4.6	4.8	4.2	4.0	3.4	3.8
Clear-Sky Solar Irradiance	taub	0.285	0.305	0.332	0.379	0.406	0.410	0.414	0.373	0.362	0.327	0.294	0.273	
	taud	2.220	2.297	2.249	2.140	2.134	2.218	2.263	2.421	2.405	2.390	2.217	2.140	
	Ebn at Noon	593	750	820	829	828	830	818	835	786	716	567	484	
	Edh at Noon	53	74	100	130	140	131	123	98	86	67	52	43	
All-Sky Solar Radiation	RadAvg	0.47	1.27	2.43	3.94	4.96	4.91	4.27	3.58	2.34	1.30	0.59	0.34	
	RadStd	0.05	0.21	0.29	0.34	0.48	0.43	0.51	0.41	0.35	0.18	0.08	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	+1.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page